STAT/MA 41600
In-Class Problem Set #34: November 6, 2026
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Problem Set 34 Answers

1. We see that $P(X \geq 3/5) = \int_{3/5}^{1} 30x^4(1-x)^1\,dx = 2396/3125 = 0.7667$.

2. We have $E(X) = \int_0^1 (x)(30x^4(1-x)^1)\,dx = 30\int_0^1 (x^5 - x^6)\,dx = 30(x^6/6 - x^7/7)|_{x=0}^1 = 5/7$.

3. We compute $P(X > U) = \int_0^1 \int_u^1 (30x^4(1-x)^1)\,dx\,du = 30\int_0^1 \int_u^1 (x^4 - x^5)\,dx\,du = 30\int_0^1 (x^5/5 - x^6/6)|_{x=u}^1\,du = 30\int_0^1 ((1/5 - 1/6) - (u^5/5 - u^6/6))\,du$ and this yields $P(X > U) = 30((1/5 - 1/6)u - (u^6/30 - u^7/42))|_{u=0}^1 = 30((1/5 - 1/6) - (1/30 - 1/42)) = 5/7$.

Alternatively, we have $P(X > U) = \int_0^1 \int_0^x (30x^4(1-x)^1)\,du\,dx = 30\int_0^1 \int_0^x (x^4 - x^5)\,du\,dx = 30\int_0^1 (x^4 - x^5)u|_{u=0}^x\,dx = 30\int_0^1 (x^5 - x^6)\,dx = 30(x^6/6 - x^7/7)|_{x=0}^1 = 30(1/6 - 1/7) = 5/7$.

4. We compute as follows: $P(X > 1/2) = \int_{1/2}^1 12x^2(1-x)^1\,dx = 12\int_{1/2}^1 (x^2 - x^3)\,dx = 12(x^3/3 - x^4/4)|_{x=1/2}^1 = 12((1/3 - 1/4) - (1/24 - 1/64)) = 11/16 = 0.6875$.

5. We compute $P(|X - 1/2| < 2/5) = P(1/10 < X < 9/10) = \int_{1/10}^{9/10} 12x^2(1-x)^1\,dx = 12\int_{1/10}^{9/10} (x^2 - x^3)\,dx = 12(x^3/3 - x^4/4)|_{x=1/10}^{9/10}$ which simplifies to $P(|X - 1/2| < 2/5) = 12((243/1000 - 6561/40000) - (1/3000 - 1/40000)) = 118/125 = 0.944$.